
Testing on Live Application

Section A: Scenario Based Conceptual Questions

1. **An online learning platform releases new courses every month. Identify the most suitable software development life cycle model and justify your choice.**

Agile model is the best choice because it allows the team to develop and release features in small parts. Since new course are added every month, agile helps in making changes quickly and delivering updates regularly.

2. **For a course purchase feature, define testing objectives from a quality assurance perspective.**

- Check that payment is completed successfully.
- Make sure duplicate payments do not happen.
- Verify that users can access the course after purchase.
- Check proper error message for failed payments.
- Ensure payment and user data are secure.

3. **Explain how non-functional testing contributes to better learner experience.**

Non-functional testing makes the platform faster, safer and easier to use. it checks performance, security, usability and compatibility on different devices and browser.

4. **During sprint execution, how should testers ensure acceptance criteria are testable?**

Tester should ensure that acceptance criteria are clear, simple, and measurable. Every requirement should be easy to convert into test case and should not create confusion.

Section B: Practical Manual Testing Tasks

1. **STLC (Student Login): Document the full lifecycle including Requirement Analysis, Planning, Test Design, Execution, and Closure. Define specific Entry and Invalid numeric test data required to verify the input limits.**

STLC : STUDENT LOGIN MODULE

Phase 1: Requirement Analysis

- Review login requirements.
- Verify valid and invalid login functionality.
- Check password reset feature.
- Check session timeout.
- Understand security and performance requirements.
- Discuss unclear requirements with the product owner.

Phase 2: Test Planning

- Define test scope: Student Login, Logout, Session Management, Password Reset.
- Identify testing type: Function, UI, Security, Performance, Regression.
- Estimate effort: 4 testers, 2days for test case writing, 1 day for execution.
- Select test environment: chrome, Firefox, Edge.
- Define risks: environment downtime, Unavailability of test data.

Phase 3: Test Design

- Write test cases covering for: valid login, invalid email format, wrong password, blank field, SQL injection, session expiry.
- Create test data: valid user account, locked account, inactive accounts.
- Apply Boundary Value Analysis and Equivalence Partitioning.
- Designing positive and negative test case

Phase 4: Test Execution

- Execute all test cases as per test plan.
- Log defects in JIRA with severity, steps to reproduce, screenshots.
- Retest fixed defects and perform regression testing.
- Track test execution progress using a traceability matrix.

Phase 5: Test Closure

- Prepare Test Summery Report. (total case, passed , failed, blocked)
- Analyse defect.
- Store testing documents. (test case, defect report, test data)
- Share lesson learned.
- Get final approval for release.

Criteria Type	Criteria
Entry Criteria	Requirement are approved.
Entry Criteria	Test environment is ready.
Entry Criteria	Build is deployed successfully.
Entry Criteria	Test cases are reviewed and approved.
Exit Criteria	Most test cases are executed.
Exit Criteria	Critical and high-priority defect are fixed.
Exit Criteria	No major defects remain open.
Exit Criteria	Test summery report is completed and approved.

2. Test Design(Course Rating 1-5): Apply Equivalence Partitioning and Boundary Value Analysis to the 1–5 star rating input. List the specific Valid and Invalid numeric test data required to verify the input limits.

Boundary Value Analysis: testing edge value like min/max.

Test value	Type	Expected result
0	Invalid (just below min)	Error: Rating must be between 1 and 5.
1	Valid (min boundary)	Ratting of 1 accepted successfully.
2	Valid (just above min)	Ratting of 2 accepted successfully.
3	Valid (normal/middle value)	Ratting of 3 accepted successfully.
4	Valid (max boundary)	Ratting of 4 accepted successfully.
5	Valid (max boundary)	Ratting of 5 accepted successfully.

Equivalence Partitioning: dividing input data into valid/invalid groups

Partition	Range	Test Value	Expected Result
Valid partition	1 to 5 (integers)	3	Rating accepted and save
Invalid partition (below)	Less than 1	0	Error: Rating must be between 1 and 5
Invalid partition (above)	Greater than 5	6	Error: Rating must be between 1 and 5
Invalid partition (non-numeric)	Letteras/special char	@\$#	Error: only numeric values allowed
Invalid partition (decimal)	Decimal number	1.11	Error: only whole number allowed

3. Test Cases (Course Experience): Write one functional test case for each: Course Search & Browse, Course Learning Experience, and Assessments & Quizzes. Each case must include Steps, Expected Result, and Pass/Fail Status.

Test Cases – Course Experience :

Field	TC-01: Search Course & Browse	TC-02: course Learning Experience	TC-03: Assessments & Quizzes
Test Case ID	TC-SRCH-01	TC-LEARN-01	TC-QUIZ-01
Module	Course Search	Learning Module	Quiz Module
Precondition	User has logged into the application.	User is logged in and has enrolled in a course.	User has finished the course chapter and quiz is available.
Steps	1. Open the search box. 2. Type "Python". 3. Click Search. 4. Check the results. 5. Select Beginner filter. 6. Open a course.	1. Open My Courses. 2. Select an enrolled course. 3. Open the first video lesson. 4. Start the video. 5. Pause and play again. 6. Move video to another time and check progress.	1. Finish the learning module. 2. Open the Assessment section. 3. Start the quiz. 4. Answer all questions. 5. Submit the quiz. 6. Check the score.
Expected Result	Beginner-level Python courses should be displayed with course details like title, trainer name, rating, and price.	Video should run smoothly. Play, pause, and seek options should work properly. Progress should update correctly.	Quiz should be submitted successfully. Score should be shown correctly and saved in the user's account.
Actual Result	Search results appeared correctly.	Video functions worked properly.	Quiz score was displayed and saved successfully.
Status	PASS	PASS	PASS

4. **Defect Log:** Document 3 unique defects found in the modules above. Each report must include: Defect Title, Severity, Steps to Reproduce, Actual vs. Expected Result, and Environment details.

DEFECT – 1 : Shopping Cart

Field	Details
Defect ID	DEF-001
Defect Title	Shopping cart is not updated after adding a product
Module	Shopping Cart
Severity	High (S2)
Priority	High (P1)
Steps to Reproduce	1. Login to the application. 2. Search for a product. 3. Click Add to Cart. 4. Open the cart page.
Actual Result	Product is added successfully, but the cart count is not updated.
Expected Result	The cart count should increase and show the added product.
Environment	Windows 10, Chrome

DEFECT – 2 : User Profile

Field	Details
Defect ID	DEF-001
Defect Title	User profile page does not open after successful login
Module	User Profile
Severity	High (S2)

Priority	High (P1)
Steps to Reproduce	1. Open the website login page. 2. Enter valid username and password. 3. Click the Login button. 4. Click on the Profile option.
Actual Result	Login is completed successfully, but when the user clicks on Profile, the page does not open and remains on the current screen.
Expected Result	After clicking the Profile option, the user's profile page should open successfully.
Environment	Windows 10 , chrome

DEFECT – 3

Field	Details
Defect ID	DEF-002
Defect Title	Invalid rating value "0" is accepted by the system
Module	Course Rating
Severity	Medium (S3)
Priority	Medium (P2)
Steps to Reproduce	1. Login with a student account. 2. Open any course that has been completed. 3. Go to the rating section. 4. Enter "0" as the rating value. 5. Click on Submit Rating.
Actual Result	The application accepts the rating value 0 and shows it as 0 stars without displaying any validation message.
Expected Result	The system should not allow rating 0 and should show a message asking the user to enter a rating between 1 and 5 stars.
Environment	OS: Android 16, Browser: Chrome

SECTION C: Mini Capstone Project – End-to-End Testing Case Study

1. **Test Planning Document: Prepare a Test Plan covering scope, objectives, testing types, environment, risks, assumptions, and entry and exit criteria.**

Section	Answer
Project Name	Online Food Delivery Application Testing
Scope	User Registration, Login, Restaurant Search, Food Ordering, Payment, Order Tracking, and Logout.
Objectives	To verify that users can order food successfully and all application features work correctly.
Testing Types	Functional Testing, UI Testing, Regression Testing, Security Testing, Performance Testing, and UAT.
Test Environment	Windows 10, Android Mobile, Chrome Browser, Firefox Browser, and Staging Environment.
Risks	Payment gateway issues, server downtime, and unavailable test data.
Assumptions	Test environment is available, internet connection is stable, and test accounts are created.
Entry Criteria	Requirements are approved, build is deployed, test environment is ready, and test cases are prepared.
Exit Criteria	All important test cases are executed, critical defects are fixed, and testing report is completed.

2. **Requirements Traceability Matrix: Create a Requirements Traceability Matrix mapping requirements to test scenarios, test cases, and defects.**

Online Shopping Application:

Requirement ID	Requirement	Test Scenario	Test Case ID	Status	Defect ID
REQ-001	User can log in with valid credentials	Verify successful login	TC-LOGIN-001	PASS	-
REQ-002	Error message should appear for invalid login	Verify login with wrong password	TC-LOGIN-002	PASS	-
REQ-003	User can search products	Search product by name	TC-SRCH-001	PASS	-
REQ-004	User can place an order	Complete product purchase	TC-ORD-001	PASS	-
REQ-005	Payment failure should show an error message	Test declined payment	TC-PAY-001	FAIL	DEF-001
REQ-006	User can view purchased orders	Check order history	TC-ORD-002	PASS	-
REQ-007	Product details should display correctly	Open product details page	TC-PROD-001	PASS	-
REQ-008	User can submit product ratings	Submit rating between 1 and 5	TC-RATE-001	FAIL	DEF-002
REQ-009	User should be redirected to dashboard after login	Verify dashboard redirection	TC-LOGIN-003	FAIL	DEF-003
REQ-010	User can log out successfully	Verify logout functionality	TC-LOGOUT-001	PASS	-

3. End-to-End Test Scenario: Write one detailed End-to-End test scenario covering: Login → Course Selection → Payment → Course Access

Step No.	Action	Test Data	Expected Result	Status
1	Open the login page	URL: learningplatform.com	Login page should open successfully	PASS
2	Enter valid login details and click Login	Email: student@test.com Password: Test@123	User should log in and open the dashboard	PASS
3	Search for a course	Keyword: Manual Testing	Relevant courses should be displayed	PASS
4	Select a course	Manual Testing Course	Course details page should open	PASS
5	Click on Enroll Now button	N/A	User should be redirected to the payment page	PASS
6	Enter payment details and	Test Card Details	Payment should be successful and confirmation	PAS

	complete payment		message should appear	
7	Open My Courses section	N/A	Purchased course should be visible in the list	PASS
8	Click on the purchased course	N/A	Course content page should open successfully	PASS
9	Start the first video lesson	N/A	Video should play without any issue	PASS
10	Verify course access	N/A	User should be able to access all purchased course content	PASS

4. Test Summary Report: Prepare a Test Summary Report including execution status, defects identified, and release recommendation.

1. Project Information

Field	Details
Project Name	Online Learning System Testing
Test Cycle	Sprint 4 / Version 2.4.1
Report Date	May 2026
Test Duration	1 June 2026 – 2o June 2026
Prepared By	QA student – TOPS Technologies

2. Test Execution Summary

Metric	Count
Total Test Cases	35
Executed Test Cases	33
Passed	29
Failed	4

Blocked	2
Not Executed	2
Pass Percentage	87.9%

3. Defect Summary

Severity	Tota	Open	Fixed	Deferred
Critical (S1)	1	0	1	0
High (S2)	2	1	1	0
Medium (S3)	1	0	1	0
Low (S4)	0	0	0	0

4. Key Defects Identified

Defect ID	Defect Description	Current Status
DEF-001	Dashboard not opening after login	Fixed
DEF-002	Rating value 0 accepted by system	Fixed
DEF-003	Quiz submission allowed after timer ends	Open
DEF-004	No error message shown when payment fails	Fixed

5. Release Recommendation

Item	Details
Decision	Conditionally Approved
Reason	Major defects are fixed and application is stable.
Condition 1	DEF-003 should be fixed in the next sprint.
Condition 2	Regression testing should be performed after the fix.
Condition 3	Monitor login and payment features after release.

6. Approval Status

Role	Status
QA Lead	Pending

Product Owner	Pending
----------------------	----------------

7. Conclusion

Result
Testing is completed successfully. Most test cases passed and critical defects are fixed. The application is ready for release with conditions.